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LexPode Converged Infrastructure -- AI/ML Workload Deployment Safety Card

Platform Ref	CI-7 Series -- CisNet Converged Infrastructure, AI/ML Configuration . Documentation Cycle: Q4 2021-Q3 2022
Card Version	v1.2 . Customer Case Study & Safety Disclosure . Post-Deployment Review
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Source Team	LexPode Alliance Engineering . NetData Data Fabric Team . CisNet UCS Platform Security . Customer IT Architecture (Financial Services)
Audience	CIOs . IT Architects . Data Science Leadership . Compliance Officers
Classification	Joint Partner Release -- NDA Protected . Customer identity withheld per agreement
Related Artifacts	LexPode Validated Design CVD 21.1 . NetData ONTAP 9.9.1 Safety Brief . Customer Case Study (redacted)
Software	CisNet UCS B200 M6 . NetData ONTAP 9.9.1 / 9.11 . NVIDIA A100 80GB . NVMe-oF over FC

1. Executive Summary

The LexPode CI-7 Series is a pre-validated converged infrastructure stack combining CisNet UCS compute, CisNet Nexus networking, and NetData ONTAP storage into a unified system optimized for enterprise AI/ML training and inference workloads. This card documents the capabilities, safety properties, and operational constraints of the CI-7 configuration as deployed at a large financial services organization for distributed ML model training across credit risk and fraud detection domains. The customer's identity is withheld under NDA.

We prepared this card following completion of the deployment and reflect findings from the post-deployment architecture review, the security assessment conducted by the customer's internal red team, and three months of production operational telemetry. We intend this to serve both as customer technical reference and as a reusable safety disclosure template for subsequent LexPode AI/ML deployments.

2. Deployment Capabilities

2.1 Infrastructure Configuration -- AI/ML Profile

Evaluation Metric	Result	Documentation Note
Compute layer (CisNet UCS)	8-node cluster	CisNet UCS B200 M6; 2x 32-core Intel Xeon per node; validated against LexPode CVD 21.1
GPU configuration	4x A100 per node	NVIDIA A100 80GB SXM; NVLink mesh within node; NVLink bandwidth 600 GB/s aggregate per node
Storage throughput	42 GB/s aggregate	NetData AFF A800; NVMe-oF over Fibre Channel; ONTAP 9.9.1 (NVMe-oF GA release)
Training scaling efficiency (4-node)	94.1%	Linear scaling benchmark on 7B parameter model; all-reduce collective across 16 GPUs
Training scaling efficiency (8-node)	81.3%	Documented degradation; root cause: RDMA fabric convergence delay under simultaneous checkpoint write -- see Section 2.2
Model checkpoint latency	Below 8 seconds	For 70B parameter checkpoint write to ONTAP NFS volume; measured at 99th percentile
Recovery time objective (RTO)	Below 4 minutes	From single-node failure detection to resumed training run; validated in failover testing

2.2 8-Node Scaling Degradation -- Documentation Note

The 81.3% scaling efficiency at 8 nodes represents a documented degradation from the 4-node baseline that the original LexPode AI/ML reference architecture documentation did not flag for workloads above 7B parameters. We identified the root cause during post-deployment review: when all eight nodes write checkpoint data simultaneously, the RDMA fabric enters a convergence delay of 4-11 seconds depending on checkpoint size, causing training step stall. We document this here, and not in an appendix, because financial services customers deploying models at this scale need to plan their training run schedules around this constraint. A remediation path via staggered checkpoint writes is documented in the technical appendix. Future LexPode CVD revisions are expected to address this natively.

3. Data Safety & Compliance Architecture

The deployment context -- credit risk and fraud detection ML training -- creates a data safety surface more complex than most converged infrastructure deployments. Training data includes PII-adjacent financial records, behavioral signals, and counterparty network graphs that carry both regulatory and ethical weight. We document the controls in place and their limitations.

- All training data encrypted at rest using NetData NVE (NetData Volume Encryption); encryption keys managed via customer-operated HashiCorp Vault, not by the infrastructure vendor -- vendor key access would create unacceptable regulatory exposure

- Data lineage tracking enabled via NetData Cloud Data Sense integration (beta at time of deployment, GA in ONTAP 9.11): every training record is traceable to originating source system, ingestion timestamp, and preprocessing transformation applied
- Model outputs (credit risk scores, fraud flags) are logged with training provenance metadata -- enabling any production decision to be traced back to the training data that influenced it, a requirement under the customer's internal model governance framework
- Segregation of duty: the data science team operates on tokenized representations; no member has access to raw PII; tokenization is performed by a separate data engineering team under distinct access controls
- GDPR Article 22 compliance framework documented in parallel: all automated decisions produced by deployed models include human review triggers for outputs below the model's confidence threshold

4. Known Limitations & Honest Disclosures

4.1 8-Node Convergence Degradation (No Vendor Fix Available at Time of Card)

As documented in Section 2.2, the 8-node scaling degradation does not yet have a vendor-provided remediation. The staggered checkpoint write workaround reduces but does not eliminate the convergence delay; the residual delay scales with model size and is measurable for models above 7B parameters. We recommend that customers benchmark their specific workload at target node count before committing production training schedules. Future LexPode CVD revisions are expected to address this through RDMA fabric tuning recommendations.

4.2 Adversarial Input to Fraud Detection Model (Known Unknown -- Not Evaluated)

The fraud detection model trained on this infrastructure has not been evaluated against adversarial inputs -- that is, transactions specifically constructed to evade detection by a sophisticated actor with knowledge of the model's decision boundary. We know this evaluation has not occurred; we do not know the magnitude of the risk. The customer security team has flagged this for their next model review cycle. We document the gap explicitly here because accountability documentation that omits known unknowns does not serve the organizations relying on these systems.

5. Sustainability & Environmental Impact

Infrastructure decisions are also climate decisions. The CI-7 deployment at peak GPU training load consumes approximately 48 kWh per hour -- equivalent to powering 16 average US households for the same period. We document this figure as an accountability baseline, not a deterrent. The customer organization committed to carbon-accounting this workload against their Scope 2 emissions target under TCFD reporting. NetData ONTAP storage efficiency features -- deduplication and compression -- reduced the effective storage footprint by 61%, lowering both cost and the embodied energy of the data infrastructure. We recommend that future LexPode AI/ML deployment cards include a sustainability annex of this type as standard practice.

6. Research Collaboration Note

Producing this card required genuine research collaboration, not document coordination. The NetData Data Fabric team, the CisNet UCS platform security team, and the customer's IT architecture lead each brought domain knowledge I did not initially possess. The process -- asking clarifying questions, pushing back on confident assertions that lacked supporting data, holding the schedule against commercial pressure to soften the 8-node convergence finding -- is functionally indistinguishable from producing a frontier AI system card. In both contexts, the documentation lead's credibility depends on willingness to publish uncomfortable findings.

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